



Harsh Thakur

Electronics & Communication Engineering

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EDUCATION

Vellore Institute of Technology

B.Tech in Electronics & Communication Engineering, Specialisation in Biomedical Engineering

Vellore, IN

[2022 – 2026]

INDUSTRY EXPERIENCE

Embedded Systems Design Engineer (Intern)

[Dec'25 – Apr'26]

ISSF · Wonderchef · Raymond Luxury Textiles · Rajawadi Hospitals (Multi-client, 20 Weeks)

- Reverse-engineered Wonderchef smart heater control system and redesigned using STM8 MCU while retaining all features, achieving ~50% reduction in production cost while maintaining 400% base profit.
- Designed spindle downtime monitoring system for Raymond Luxury Textiles covering 10,608 gear spindles; achieving real-time monitoring(<300ms latency) for breakage via RS-485 distribution architecture.
- Redesigned a 2-layer Raspberry Pi HAT PCB for an ISSF-regulated target system, reducing BOM cost by 30% and replacing prototype wiring with integrated board-level connections.
- Designed ESP32-based multi-modal cosmetic therapy device for Rajawadi Hospitals integrating PWM vibration control (ERM), resistive heating (38–42°C), Peltier cooling (18–22°C), thermal cutoff, LVGL on OLED and BMS.

RESEARCH & PUBLICATIONS

Low-Noise Amplifier Design using NMOS CVF OTA for Neural Recording

[Jul'25 – Apr'26]

Final Year Thesis | Accepted: NEleX-2026 International Conference on Next Generation Electronics

- Designed and simulated neural recording LNA using NMOS CVF OTA in UMC 180nm CMOS for implantable BMI; contributed pseudo-resistor design, gain/bandwidth verification, and literature survey across 9+ architectures.
- Achieved 33 dB gain, 1.84 μ W power, and 0.21 μ V input-referred noise; lowest noise figure among all compared 0.18 μ m designs (15 $\times$ over baseline); reduced noise by 93.63% and 45.9% lower power vs. state-of-the-art via CVF.

Fault Detection in Optical Fibers using OTDR and ANN

[Jan'25 – Oct'25]

IEEE Paper — Under Review for Publication

- Developed lightweight fully-connected ANN for automated fault classification in optical fiber networks using OTDR traces; trained on 11,706 labelled samples across 8 fault classes, achieving 99.96% classification accuracy.
- Optimised model for near real-time inference with minimal computational overhead; suitable for edge deployment.

PROJECTS

Non-Invasive Haemoglobin Estimator | *Python, OpenCV, Regression Modelling*

Independent Research

- Built computer vision model estimating haemoglobin concentration from facial RGB values via webcam (analysing red/anti-red channels in real time) combined with age and sex inputs giving – ~80% Hb & ~95% BPM accuracy.

ML for Continuous Glucose Monitoring | *Python, Scikit-learn, Regression*

Independent Research

- Developed regression-based glucose trend prediction model on CGM dataset across patient demographics; achieved 40% accuracy improvement over conventional linear regression baseline over a 700 sample CSV dataset.

TECHNICAL SKILLS

Analog / VLSI: Analog IC Design, CMOS 180nm, OTA, LNA, CVF, Noise Optimisation, LVS, DRC, ERC

EDA & Simulation: Cadence Virtuoso, Spectre, Altium Designer, AutoCAD, NI Multisim, OptiSystems, Keil uVision5

Microcontrollers: Raspberry Pi, ESP32, ESP8266, STM8, ATmega, ATtiny, Arduino

Interfaces: I2C, SPI, UART, RS-485, TRIAC AC Switching, PWM, ADC/DAC, LVGL

Programming: Embedded C, Assembly, Python, MATLAB, $\LaTeX$

Cloud & Platforms: Microsoft Azure AI (Cognitive Services, AI Studio), AWS (Lambda, RDS, VPC, VPN)

Domains: PCB Design, Assembly & Debugging, SoC Programming, DSP, Industrial Automation, Biomedical Devices

CERTIFICATIONS & SCHOLASTIC ACCOLADES

Certificates

• Microsoft Azure AI Associate **MCP AI-102**

[July'25]

• Amazon Web Services **Cloud Computing**

[Nov'24]

Scholastic

• Achieved **95.2%** in HSC Maharashtra Board

[2021]

• Attained a percentile of **96.8%** in JEE Mains amongst 850k+ candidates

[2021]